A Study in Ind-Coherent Sheaves and Correspondences

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Chapter 1

Motivation

1.1 Ind-coherent sheaves

An important method to study modern representation theory is algebro-geometric. Specifically, For linear algebraic group G, **the category of algebraic representation** $\mathrm{rep}(G)$ **can be interpreted as the category of quasi-coherent sheaves on the classifying stack** BG**.** In this view, many useful functors or operations on G are naturally correspond to the sheaf level. Explicit formulas do not work well when modern representation theory are inevitably involved in higher category.

The most significant mathematic object in this book is Ind-coherent sheaves. Before we define it, we should know the reason to drop the Quasi-coherent sheaves that Lurie used in his higher algebra.

To see the reason, let $\mathfrak{g}$ be the Lie algebra of G with associated formal group $\exp(\mathfrak{g})$. Interpret $\mathfrak{g}$-mod as modules over $U(\mathfrak{g})$ and $\mathrm{rep}(G)$ as modules of compactly supported distributions $\mathcal{E}'(G)$. In this sense, The restriction functor

$$\mathrm{rep}(G) \to \mathfrak{g}\text{-mod}$$

can be interpreted as the pullback functor along the map

$$B(\exp(\mathfrak{g})) \to BG$$

Since one can prove that the category $\mathfrak{g}$-mod is canonically equivalent to $\mathrm{QCoh}(B(\exp(\mathfrak{g})))$. Let us now be given a homomorphism $\alpha : \mathfrak{g}' \to \mathfrak{g}$. The functor of restriction $\mathfrak{g}$-mod $\to \mathfrak{g}'$-mod is the pullback functor of

$$f_\alpha : B(\exp(\mathfrak{g}')) \to B(\exp(\mathfrak{g})). \tag{$\bigstar$}$$

However the left adjoint to restriction functor is induction functor which cannot correspond to the direct image along ($\bigstar$). More explicitly speaking, in the sense of representation theory, the direct image functor as the right adjoint of pullback functor of ($\bigstar$) is coinduction functor. In the case of infinite dimension, **the Frobenius Reciprocity Ind $\cong$ Coind fails to hold and thus the geometric realization of induction functor fades.**

The birth of ind-coherent sheaves is aim to deal with the situation that Frobenius Reciprocity fails. For the case of Frobenius Reciprocity holds, for example classical Lie algebra[1], the ind-coherent sheaves are equivalent to quasi-coherent sheaves.

[1]Here, "classical" means finite dimensional

In the sense of ind-coherent sheaves, the restriction functor along the homomorphism α is given by the !-pullback functor

$$(f_\alpha)^! : \mathrm{IndCoh}(B(\exp(\mathfrak{g}))) \to \mathrm{IndCoh}(B(\exp(\mathfrak{g}')))$$

the functor of induction $\mathfrak{g}'$-mod $\to$ $\mathfrak{g}$-mod will be given by the functor of IndCoh direct image

$$(f_\alpha)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(B(\exp(\mathfrak{g}'))) \to \mathrm{IndCoh}(B(\exp(\mathfrak{g}))).$$

An analytical analogy is QCoh **behaves more like functions on a space, while** IndCoh **behaves more like measures.**

For every scheme of finite type X, its de Rham prestack X_{dR} is a inf-scheme, one can prove that

$$\mathrm{IndCoh}(X_{dR}) \cong \text{D-mod}(X)$$

Then for an arbitrary inf-scheme $\mathcal{X}$, the ind-coherent sheaf can be given by

$$\mathrm{IndCoh}(\mathcal{X}) = \lim_{Z \to \mathcal{X}} \mathrm{IndCoh}(Z).$$

Hence, one can prove that for any proper map $f : \mathcal{X}' \to \mathcal{X}$, the pair of functor $(f_*^{\mathrm{IndCoh}}, f^!)$ is adjoint. There are several examples to illustrate its advantages.

Example 1.1.1. Take $f : X \to X_{\mathrm{dR}}$ be canonical projection, then the adjoint pair identifies with the pair

$$\mathbf{ind}_{\text{D-mod}} : \mathrm{IndCoh}(X) \rightleftarrows \text{D-mod}(X) : \mathbf{ind}_{\text{D-mod}}.$$

Example 1.1.2. Take $f : Y_{\mathrm{dR}} \to X_{\mathrm{dR}}$, then f_*^{IndCoh} and $f^!$ are correspond to the functors

$$g_{*,\text{D-mod}} : \text{D-mod}(Y) \to \text{D-mod}(X) \quad \text{and} \quad g^!_{\text{D-mod}} : \text{D-mod}(X) \to \text{D-mod}(Y)$$

of D-module push-forward and pullback respectively.

Example 1.1.3. Consider Lie algebra $\mathfrak{g}$ acts on a scheme X. In this case, we have a diagram

$$B(\exp(\mathfrak{g})) \xleftarrow{f_1} B_X(\exp(\mathfrak{g})) \xrightarrow{f_2} X_{\mathrm{dR}}.$$

Then the composite functor

$$(f_2)_*^{\mathrm{IndCoh}} \circ (f_1)^! : \mathrm{IndCoh}(B(\exp(\mathfrak{g}))) \to \mathrm{IndCoh}(X_{\mathrm{dR}})$$

identifies with the localization functor $\mathfrak{g}$-mod $\to$ D-mod(X).

Remark 1.1.4. BG manifests as a geometric representable functor

$$BG : \mathrm{dSch}^{\mathrm{op}} \to \infty\text{-Grpd} \quad X \mapsto \mathrm{Bun}_G(X).$$

From an operadic perspective,

$$\mathrm{Bun}_G(X) \simeq \mathrm{Alg}_X(BG) \simeq \mathrm{Map}_{\mathrm{PreStk}}(X, BG)$$

where we interpret X as an unary ∞-operad and $\mathrm{Bun}_G(X)$ as the category of BG-valued algebras over X_{dR}. When we consider the de Rham prestack X_{dR}, we obtain that

$$\mathrm{LocSys}_G(X) \simeq \mathrm{Alg}_{X_{\mathrm{dR}}}(BG) \simeq \mathrm{Map}_{\mathrm{PreStk}}(X_{\mathrm{dR}}, BG).$$

$B_X(G)$ is the ∞-categorical relative tensor product $X \otimes_G *$.

1.2 Construction of ind-coherent sheaves

There is a heuristic formula we have presented

$$\mathrm{IndCoh}(\mathcal{X}) = \lim_{Z \to \mathcal{X}} \mathrm{IndCoh}(Z) \tag{$\bigstar$}$$

In what sense is this limit taken? In triangulated categories, colimits and limits are not well-behaved.

We do it in the ∞-category of presentable k-linear stable ∞-categories, or equivalently, in the ∞-category of differential graded categories: categories enriched over the category of chain complexes of k-modules, such that the differential satisfies the Leibniz rule with respect to composition. We denote it by DGCat.

For a scheme Z, the category $\mathrm{IndCoh}(Z)$ is the renormalization of $\mathrm{QCoh}(Z)$. One can prove that ind-coherent sheaves have the following properties:

a. DG category $\mathrm{IndCoh}(Z)$ contains all infinite coproducts

b. For a map $Z' \xrightarrow{f} Z$, the functor $\mathrm{IndCoh}(Z') \xrightarrow{f^!} \mathrm{IndCoh}(Z)$ preserves all infinite coproducts.

We can see that the shriek pullback functor $f^{!,\mathrm{QCoh}}$ fails to satisfy property (b) even in the case that f is proper. This is because the direct image functor f_* as the left adjoint of $f^{!,\mathrm{QCoh}}$ does not preserve compactness, and thus its right adjoint does not preserve infinite coproducts according to Brown representability. Indeed, f_* does not necessarily send perfect complexes on Z' to perfect complexes on Z unless f is of finite Tor-dimension. An easy example is $f : \mathrm{Spec}(k) \to \mathrm{Spec}(k[t]/t^2)$. The perfect complex k does not has finit resolution as a $k[t]/t^2$-module.

Fortunately, when f is proper, the direct image functor f_* sends the category of bounded coherent complexes $\mathrm{Coh}^b(Z')$ to $\mathrm{Coh}^b(Z)$ and we define ind-coherent sheaves as the ind-completion of $\mathrm{Coh}^b(-)$

$$\mathrm{IndCoh}(-) := \mathrm{Ind}(\mathrm{Coh}^b(-))$$

The compact objects in the category $\mathrm{IndCoh}(Z)$ are exactly all bounded coherent complexes, and therefore f_* is well-behaved.

The difference of QCoh and IndCoh is quite small to some extend. More clearly, the tautological embedding $\mathrm{Coh}(Z) \hookrightarrow \mathrm{QCoh}(Z)$ induces

$$\Psi_Z : \mathrm{IndCoh}(Z) \to \mathrm{QCoh}(Z)$$

and the ususal t-structure on $\mathrm{Coh}(Z)$ induces one on $\mathrm{IndCoh}(Z)$. One can prove that Ψ_Z is t-exact resulting

$$\mathrm{IndCoh}(Z)^{\geq -n} \to \mathrm{QCoh}(Z)^{\geq -n}$$

is an equivalent. Therefore, the difference occure only at $-\infty$. The difference of f_*^{IndCoh} and f_* is also small. Explicitly, f_*^{IndCoh} is the unique factor that perserves infinite direct sums and makes the diagram

$$\begin{tikzcd}
\mathrm{IndCoh}(Z')^{\geq -n} \arrow[r, "\Psi_{Z'}", "\sim"'] \arrow[d, "f_*^{\mathrm{IndCoh}}"'] & \mathrm{QCoh}(Z')^{\geq -n} \arrow[d, "f_*"] \\
\mathrm{IndCoh}(Z)^{\geq -n} \arrow[r, "\Psi_Z", "\sim"'] & \mathrm{QCoh}(Z)^{\geq -n}
\end{tikzcd}$$

commutes for every n.

The main object of this book is the functor

$$\mathrm{IndCoh}^!_{\mathrm{Sch}} : (\mathrm{Sch})^{\mathrm{op}} \to \mathrm{DGCat}. \tag{$\mathbf{\Pi}$}$$

The construction of it as follows. For a morphism $f : Z' \to Z$, we can factor it as

$$Z' \xrightarrow{f_1} \overline{Z'} \xrightarrow{f_2} Z$$

where f_1 is an open embedding and f_2 is proper. Then we can define

$$f^! := f_1^! \circ f_2^! : \mathrm{IndCoh}(Z) \to \mathrm{IndCoh}(Z')$$

where $f_2^!$ is the right adjoint of $(f_2)_*^{\mathrm{IndCoh}}$ and $f_1^!$ is the left adjoint of $(f_1)_*^{\mathrm{IndCoh}}$.

Since we are discuss in the sense of infinite categories, we need to supply the assignment

$$f \rightsquigarrow f^!$$

with a homotopy-coherent system of compatibilities for n-fold composition of morphism. In a word, we need to show the functoriality in the sense of ∞-categories.

For example, we need to consider the compatibility betweend functor ($\mathbf{\Pi}$) and functor

$$\mathrm{IndCoh} : \mathrm{Sch} \to \mathrm{DGCat} \quad Z \rightsquigarrow \mathrm{IndCoh}(Z) \quad (Z' \xrightarrow{f} Z) \rightsquigarrow (\mathrm{IndCoh}(Z') \xrightarrow{f_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}(Z)).$$

Namely, we need to show whenever we have a Cartesian square

$$\begin{tikzcd} Z_1' \arrow[r, "g'"] \arrow[d, "f_1"'] & Z' \arrow[d, "f"] \\ Z_1 \arrow[r, "g"'] & Z \end{tikzcd}$$

There is a canonical isomorphsim of functors, called base change:

$$(f_1)_*^{\mathrm{IndCoh}} \circ (g')^! \cong g^! \circ f_*^{\mathrm{IndCoh}}.$$

The appearance of the Cartesian square make us inevitably use ∞-categories. For example, consider the diagram

$$\begin{tikzcd} \mathrm{pt} \times_{\mathbb{A}^1} \mathrm{pt} \arrow[r] \arrow[d] & \mathrm{pt} \arrow[d] \\ \mathrm{pt} \arrow[r] & \mathbb{A}^1 \end{tikzcd}$$

the fiber product $\mathrm{pt} \times_{\mathbb{A}^1} \mathrm{pt}$ cannot be taken to be the point-scheme. Rather it makes sense in the derived setting:

$$k \otimes_{k[t]} k = k[\varepsilon] \quad \deg(\varepsilon) = -1.$$

Chapter 2

Derived Algebraic Geometry

Let us review concepts from Derived Algebraic Geometry.
1. 1-Cat denote the $(\infty, 1)$-category of $(\infty, 1)$-categories.
2. Spc denote the $(\infty, 1)$-category of spaces. There is a canonical faithful embedding

$$\mathrm{Set} \hookrightarrow \mathrm{Spc}$$

with a left adjoint

$$\mathcal{S} \mapsto \pi_0(\mathcal{S}).$$

3. Spc also can be defined by the ∞-category of ∞-groupoids. Since the singular functor

$$X \rightsquigarrow \mathrm{Sing}(X)$$

assigns an ∞-groupoid $\mathrm{Sing}(X)$ to a topological space X. However, it kills non-homotopic information. Formally, we can see that the inverse image functor

$$\mathrm{Psh}(\mathrm{Sing}(X)) \to \mathrm{Psh}(\mathrm{Open}(X))$$

is not an equivalence in ∞-categories. Indeed, $\mathrm{Psh}(\mathrm{Sing}(X))$ is the locally constant ∞-sheaves on X. Therefore, one can refer to Spc as Anima to imply the difference with topological spaces.
4. $\mathrm{Spc} \hookrightarrow$ 1-Cat is a full subcategory with a right adjoint

$$\mathbf{C} \mapsto \mathbf{C}^{\mathrm{Spc}},$$

which discarding non-invertible morphisms.
5. For $\mathbf{c}_0, \mathbf{c}_1$ are objects of $\mathbf{C} \in$ 1-Cat. The mapping space denoted by $\mathrm{Maps}_{\mathbf{C}}(\mathbf{c}_0, \mathbf{c}_1) \in \mathrm{Spc}$.
6. Let $\mathbf{C}^{\mathrm{ordn}}$ denotes the homotopy category $\mathrm{Ho}(\mathbf{C})$. Let $\text{1-Cat}_{\mathrm{ordn}}$ form by ordinary categories.
7. a map $\phi : \mathbf{c}_0 \to \mathbf{c}_1$ is said to be an isomorphism if $\pi_0(\phi)$ is an isomorphism.
8. Let $\mathrm{Funct}(\mathbf{D}, \mathbf{C})$ denotes $(\infty, 1)$-category of functors. We have

$$\mathrm{Funct}(*, \mathbf{C}) \cong \mathbf{C}$$

and

$$\mathrm{Maps}_{\text{1-Cat}}(\mathbf{D}, \mathbf{C}) = \mathrm{Funct}(\mathbf{D}, \mathbf{C})^{\mathrm{Spc}}.$$

9. A functor $F : \mathbf{C} \to \mathbf{D}$ is said to be an equivalence if it induces an isomorphism in $(\text{1-Cat})^{\mathrm{ordn}}$. Equivalently, we also can say F is an isomorphism of 1-Cat.

10. For a diagram of categories

$$\mathbf{C}' \longrightarrow \mathbf{C} \longleftarrow \mathbf{C}'',$$

we can form their fiber product $\mathbf{C}' \times_{\mathbf{C}} \mathbf{C}'' \in$ 1-Cat. Then we can give a definition of $\mathbf{C}^{\mathrm{Spc}}$ by fiber product

$$\mathbf{C}^{\mathrm{Spc}} = \mathbf{C} \times_{\mathbf{C}^{\mathrm{ordn}}} \mathbf{C}^{\mathrm{ordn},\simeq}$$

where $\simeq$ means dicarding all non invertible 1-morphisms.
11. The mapping space of ∞-category $\mathrm{Funct}(\mathbf{D}, \mathbf{C})$ also can be defined as

$$\mathrm{Maps}_{\mathrm{Funct}(\mathbf{D},\mathbf{C})}(F, G) = \mathbf{C}^{\mathbf{D}\times\Delta^1} \times_{(\mathbf{C}^{\mathbf{D}}\times\mathbf{C}^{\mathbf{D}})} \Delta^0$$

12. For $\mathcal{S} \in \mathrm{Spc}$ with fixed point $s \in \mathcal{S}$, we define its loop space $\Omega\mathcal{S} := * \times_{\mathcal{S}} *$ and homotopy groups $\pi_i(\mathcal{S}, s) = \pi_{i-1}(\Omega(\mathcal{S}))$.
13. 1-Cat carries a canonical involutive auto-equivalence $\mathbf{C} \to \mathbf{C}^{\mathrm{op}}$
14. Let $[n]$ denote the category $\{0 \to 1 \to \cdots \to n\}$. Let $[0]$ denote $*$.
15. Let Δ denote the full subcategory of 1-$\mathrm{Cat}_{\mathrm{ordn}}$ spanned by the objects $[n]$. Then it carries a canonical involutive auto-equivalence

$$\mathrm{rev} : i \mapsto n - i.$$

16. $F : \mathbf{D} \to \mathbf{C}$ is said to be fully faithful if for every $\mathbf{d}_1, \mathbf{d}_2 \in \mathbf{D}$ the maps

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2) \to \mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(\mathbf{d}_2))$$

is an isomorphism in Spc.
17. A map of spaces $F : \mathcal{S}_0 \to \mathcal{S}_1$ is said to be monomorphism if F is fully faithful as a functor. Equivalently, if $\pi_0(F)$ is injective and for every point $s_0 \in \mathcal{S}_0$, the induced map

$$\pi_i(\mathcal{S}_0, s_0) \to \pi_i(\mathcal{S}_1, F(s_0))$$

is an isomorphism for all $i > 0$.
18. For every full subcategory $\overline{\mathbf{C}'}$ of $\mathbf{C}^{\mathrm{ordn}}$, one can attach an ∞-category $\mathbf{C}'$ such that it is a full subcategory of $\mathbf{C}$.
19. A fully faithful functor is an equivalence onto a full subcategory.
20. A full subspace of $\mathcal{S}$ is the same as a full subcategory of $\mathcal{S}$, which bijective to subsets of $\pi_0(\mathcal{S})$. A connected component of $\mathcal{S}$ is a full subspace.
21. A functor $F : \mathbf{D} \to \mathbf{C}$ is called 1-fully faithful if

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2) \to \mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(\mathbf{d}_2))$$

is a monomorphism in Spc. If $\mathbf{C}, \mathbf{D} \in$ 1-$\mathrm{Cat}_{\mathrm{ordn}}$, then F induces an injection on Hom sets.
22. A 1-fully faithful functor $F : \mathbf{D} \to \mathbf{C}$ is said to be 1-replete if the connected components of $\mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(\mathbf{d}_2))$ that correspond to isomorphisms are the image of $\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2)$. Equivalently, if $\mathbf{D}^{\mathrm{Spc}} \to \mathbf{C}^{\mathrm{Spc}}$ is a monomorphism.
23. Let $\overline{\mathbf{C}}$ be an ordinary category. a subcategory $\overline{\mathbf{C}}'$ is said to be a 1-full subcategory if $\mathrm{Hom}_{\overline{\mathbf{C}}'}(\mathbf{c}_1, \mathbf{c}_2)$ contains all isomorphism and is closed under compositions. Equivalently, the functor $\overline{\mathbf{C}}' \to \overline{\mathbf{C}}$ is 1-replete.
24. a subcategory $\mathbf{C}'$ of $\mathbf{C}$ is said to be 1-full subcategory if thers exists a 1-full subcategory $\overline{\mathbf{C}}'$ of $\mathbf{C}^{\mathrm{ordn}}$ such that shares same objects and

$$\mathrm{Maps}_{\mathbf{C}'}(\mathbf{c}_1, \mathbf{c}_2) = \mathrm{Maps}_{\mathbf{C}}(\mathbf{c}_1, \mathbf{c}_2) \times_{\mathrm{Hom}_{\mathbf{C}^{\mathrm{ordn}}}(\mathbf{c}_1,\mathbf{c}_2)} \mathrm{Hom}_{\overline{\mathbf{C}}'}(\mathbf{c}_1, \mathbf{c}_2).$$

25. $\mathbf{C}^{\mathrm{Spc}}$ is the minimal 1-full subcategory of $\mathbf{C}$ such that holds all objects in $\mathbf{C}$.
26. Concepts above also can be defined by fiber product. A functor $F : \mathbf{D} \to \mathbf{C}$ is said to be 1-fully faithful if

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2) \times_{\mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(\mathbf{d}_2))} \mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2) \simeq \mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, \mathbf{d}_2);$$

is said to be 1-replete if

$$\mathbf{D} \simeq \mathbf{C} \times_{\mathbf{C}^{\mathrm{ordn}}} \mathbf{D}^{\mathrm{ordn}}.$$

27. In the above situation, for any $\mathbf{D} \in$ 1-Cat, the functor

$$\mathrm{Funct}(\mathbf{D}, \mathbf{C}') \to \mathrm{Funct}(\mathbf{D}, \mathbf{C})$$

is 1-replete. Since functor $\mathrm{Funct}(\mathbf{D}, -)$, as the right adjoint of functor $- \times \mathbf{D}$, keeps pullback. Therefore, if a functor $\mathbf{D} \to \mathbf{C}$ can be factored through $\mathbf{C}'$, it can be done in an essentially unique way. One can also prove this claim by the fact that

$$\mathrm{Maps}_{\mathrm{Funct}(\mathbf{D},\mathbf{C})}(F, G) \simeq \lim_{(c \to d) \in \mathrm{Tw}(\mathbf{D})} \mathrm{Maps}_{\mathbf{C}}(F(c), G(d))$$

28. Inherite notation above. If functor $F : \mathbf{D} \to \mathbf{C}$ corresponds to functor $\mathbf{D}^{\mathrm{ordn}} \to \mathbf{C}^{\mathrm{ordn}}$ factors through $\overline{\mathbf{C}'}$, then F gives rise to a well-defined functor $\mathbf{D} \to \overline{\mathbf{C}'}$. The functor $\mathbf{D} \to \mathbf{C}'$ is an equivalence in and only if F is 1-replete and $\mathbf{D}^{\mathrm{ordn}} \to \overline{C}'$ is an equivalence.
29. A functor $F : \mathbf{D} \to \mathbf{C}$ is said to be conservative if for an $\alpha \in \mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_0, \mathbf{d}_1)$ the fact that $F(\alpha)$ is an isomorphism implies that α is an isomorphism. Equivalently, if

$$\mathbf{D}^{\mathrm{Spc}} \simeq \mathbf{D} \times_{\mathbf{C}} \mathbf{C}^{\mathrm{Spc}}.$$

30. Let $F : \mathbf{D} \to \mathbf{C}$ be a functor. we call a morphism $\mathbf{d}_0 \xrightarrow{\alpha} \mathbf{d}_1$ is Cartesian over $\mathbf{C}$ if for every $\mathbf{d}' \in \mathbf{D}$, we have

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}', \mathbf{d}_0) \to \mathrm{Maps}_{\mathbf{D}}(\mathbf{d}', \mathbf{d}_1) \times_{\mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}'), F(\mathbf{d}_1))} \mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}'), F(\mathbf{d}_0))$$

is an isomorphism.
31. Inherite notation above. Inteprete $\alpha : \mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0) \to \mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_1)$ as a morphism in $\mathrm{Psh}(\mathbf{D})$. We call α is Cartesian if

$$\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0) \simeq \mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_1) \times_{\mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_1))} \mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_0))$$

Here, for example $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_1)$ is regarded as a object in $\mathrm{Psh}(\mathbf{D})$. But by Grothendieck's construction it can be geometric realized to slice category $\mathbf{D}_{/\mathbf{d}_1}$, then fiber product is well-defined.
32. Inherite notation above again. We can identify

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}', \mathbf{d}_0) \simeq \mathrm{Tw}(\mathbf{D}) \times_{(\mathbf{D}^{\mathrm{op}} \times \mathbf{D})} (\mathbf{d}', \mathbf{d}_0).$$

We also can write

$$\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0) \simeq \mathrm{Tw}(\mathbf{D}) \times_{\mathbf{D}} \mathbf{d}_0.$$

It means that presheaf $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0)$ is a fiber bundle on $\mathbf{d}_0$. Then α is Cartesian over $*$ if the pullback of fiber bundle $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_1)$ over $\mathbf{d}_1$ along α is equivalent to $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0)$. In the relative case, the relative fiber bundle on $\mathbf{d}_0$ is

$$\mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_0)) \simeq \mathrm{Tw}(\mathbf{C}) \times_{\mathbf{C}} \mathbf{d}_0.$$

Then α is Cartesian over $\mathbf{C}$ if fiber bundle $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_1)$ pullback along

$$\mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_0)) \to \mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_1))$$

is equivalent to fiber bundle $\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0)$ on $\mathbf{d}_0$.

33. A functor $F : \mathbf{D} \to \mathbf{C}$ is said to be a Cartesian fibration if for every morphism $\mathbf{c}_0 \to \mathbf{c}_1$ and $\mathbf{d}_1$ such that $F(\mathbf{d}_1) \simeq \mathbf{c}_1$ there exists a Cartesian morphism $\mathbf{d}_0 \to \mathbf{d}_1$ that fits into a commutative diagram

$$\begin{tikzcd} F(\mathbf{d}_0) \arrow[r] \arrow[d, "\sim"'] & F(\mathbf{d}_1) \arrow[d, "\sim"] \\ \mathbf{c}_0 \arrow[r] & \mathbf{c}_1 \end{tikzcd}$$

We also call $\mathbf{d}_0 \to \mathbf{d}_1$ the Cartesian lifting of $\mathbf{c}_0 \to \mathbf{c}_1$. A functor is a Cartesian fibration if it admits Cartesian lifts for all morphisms in the base category.

34. Inherite notation above. We say F is a Cartesian fibration in spaces if it is a Cartesian fibration and every

$$\mathbf{D}_{\mathbf{c}} := \mathbf{D} \times_{\mathbf{C}} \mathbf{c}$$

is a space. We also call F a right fibration if F is a Cartesian fibration in spaces.

35. We can see that $F : \mathbf{D} \to \mathbf{C}$ is a Cartesian fibration if and only if functor

$$\mathrm{Tw}(\mathbf{D}) \to \mathrm{Tw}(\mathbf{C}) \times_{\mathbf{C}} \mathbf{D} \tag{$\bigstar$}$$

is a right fibration. To see that, we need to note that F is Cartesian if and only if

$$\mathrm{Maps}_{\mathbf{D}}(-, \mathbf{d}_0) \simeq \mathrm{Maps}_{\mathbf{C}}(F(-), F(\mathbf{d}_0)) \times_{(\mathrm{Tw}(\mathbf{C}) \times_{\mathbf{C}} \mathbf{D})} \mathrm{Tw}(\mathbf{D}).$$

One can prove that if $p : \mathcal{E} \to \mathcal{B}$ is a Right fibration if and only if

$$\mathcal{E}_{/x} \longrightarrow \mathcal{E} \times_{\mathcal{B}} \mathcal{B}_{/p(x)}$$

is an equivalence. Then prove ($\bigstar$) is trivial.

36. We call $F : \mathbf{D} \to \mathbf{C}$ a Cocartesian fibration if

$$\mathrm{Tw}(\mathbf{D}) \to \mathbf{D} \times_{\mathbf{C}} \mathrm{Tw}(\mathbf{C}) \tag{$\mathbf{II}$}$$

is a left fibration. Equivalently, if

$$\mathrm{Maps}_{\mathbf{D}}(\mathbf{d}_1, -) \simeq \mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(-)) \times_{\mathbf{D} \times_{\mathbf{C}} \mathrm{Tw}(\mathbf{C})} \mathrm{Tw}(\mathbf{D}).$$

In this case, the relative fiber bundle on $\mathbf{d}_1$ is a precosheaf

$$\mathrm{Maps}_{\mathbf{C}}(F(\mathbf{d}_1), F(-))$$

which is the pullback of ($\mathbf{II}$) along $\mathbf{d}_1 \to \mathbf{D} \times_{\mathbf{C}} \mathrm{Tw}(\mathbf{C})$.

37. Given a functor $F : \mathbf{I} \to \mathbf{C}$ we define over-category

$$\mathbf{C}_{/F} := \mathbf{C} \times_{\mathrm{Funct}(\mathbf{I},\mathbf{C})} \mathrm{Funct}([1] \times \mathbf{I}, \mathbf{C}) \times_{\mathrm{Funct}(\mathbf{I},\mathbf{C})} \{F\}$$

and under-category

$$\mathbf{C}_{F/} := \{F\} \times_{\mathrm{Funct}(\mathbf{I},\mathbf{C})} \mathrm{Funct}([1] \times \mathbf{I}, \mathbf{C}) \times_{\mathrm{Funct}(\mathbf{I},\mathbf{C})} \mathbf{C}.$$

If we take $\mathbf{I} = *$ and F gives an object in $\mathbf{C}$, then we simply write $\mathbf{C}_{/\mathbf{c}}$ and $\mathbf{C}_{\mathbf{c}/}$.
38. The forgetful functors

$$\mathbf{C}_{/F} \to \mathbf{C} \text{ and } \mathbf{C}_{F/} \to \mathbf{C}$$

is right fibration and left fibration.
39. Indeed we have

$$\mathrm{Maps}_{\mathbf{C}}(\mathbf{c}_0, \mathbf{c}_1) \simeq \mathrm{Funct}([1], \mathbf{C}) \times_{\mathbf{C}\times\mathbf{C}} (\mathbf{c}_0, \mathbf{c}_1)$$

and thus

$$\mathbf{C}_{/\mathbf{c}} \simeq \mathrm{Maps}_{\mathbf{C}}(-, \mathbf{c}) \text{ and } \mathbf{C}_{\mathbf{c}/} \simeq \mathrm{Maps}_{\mathbf{C}}(\mathbf{c}, -)$$

with fiber

$$(\mathbf{C}_{/\mathbf{c}})_{\mathbf{c}'} \simeq \mathrm{Maps}_{\mathbf{C}}(\mathbf{c}', \mathbf{c}) \text{ and } (\mathbf{C}_{\mathbf{c}/})_{\mathbf{c}'} \simeq \mathrm{Maps}_{\mathbf{C}}(\mathbf{c}, \mathbf{c}')$$

40. Given by a functor $F : [1] \to \mathbf{C}$ such that $F(i) = \mathbf{c}_i$, we let $\mathbf{C}_{\mathbf{c}_0//\mathbf{c}_1}$ denotes

$$\mathrm{Funct}([2], \mathbf{C}) \times_{\mathrm{Funct}([1],\mathbf{C})} \{F\}$$

where $\mathrm{Funct}([2], \mathbf{C}) \to \mathrm{Funct}([1], \mathbf{C})$ is induced by coface $\delta_1 : [1] \to [2]$.
41. Fix a ∞-category $\mathbf{C}$. Consider the category $1\text{-Cat}_{/\mathbf{C}}$ whose objects are pairs $(\mathbf{D}, F : \mathbf{D} \to \mathbf{C})$. Let $\mathrm{coCart}_{/\mathbf{C}}$ (resp. $0\text{-coCart}_{/\mathbf{C}}$) be the full subcategory of $1\text{-Cat}_{/\mathbf{C}}$ whose objects are those $(\mathbf{D}, F)$, for F is a coCartesian fibration (resp. left fibration). Let $(\mathrm{coCart}_{/\mathbf{C}})_{\mathrm{strict}}$ be the 1-full subcategory of $\mathrm{coCart}_{/\mathbf{C}}$, where we allow as 1-morphisms those functors $\mathbf{D}_1 \to \mathbf{D}_2$ over $\mathbf{C}$ that send coCartesian arrows to coCartesian arrows.
42. Unstraightening
43. There is a canonical equivalence between $(\mathrm{coCart}_{/\mathbf{C}})_{\mathrm{strict}}$ and $\mathrm{Funct}(\mathbf{C}, 1\text{-Cat})$. Under this equivalence, the full subcategory

$$0\text{-coCart}_{/\mathbf{C}} \subset (\mathrm{coCart}_{/\mathbf{C}})_{\mathrm{strict}}$$

corresponds to the full subcategory

$$\mathrm{Funct}(\mathbf{C}, \mathrm{Spc}) \subset \mathrm{Funct}(\mathbf{C}, 1\text{-Cat}).$$